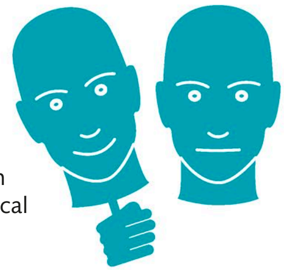
Altruism

Altruism—when a person acts to benefit another at personal cost or risk—involves empathizing with another's distress then acting to help. It involves distinct processes. Brain scans show that acting altruistically activates the reward pathways (see pp.112–113), reinforcing the behavior and quelling emotional discomfort. Selflessness is a distinguishing feature of human behavior and an evolutionary enigma given dangers to the altruist.

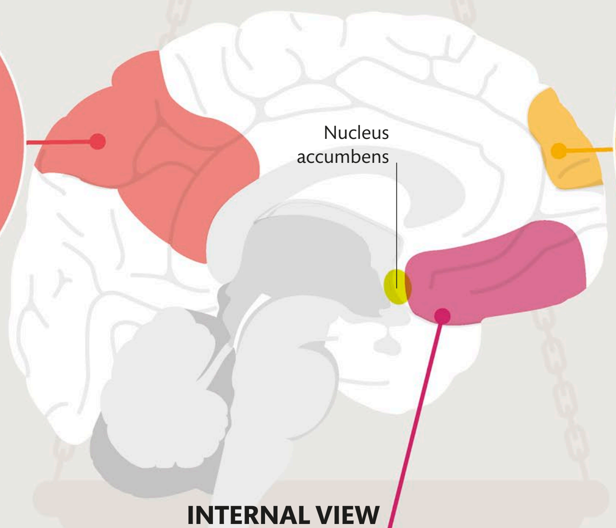
PSYCHOPATHY

Psychopaths can understand morality and can, therefore, mimic normal social interactions. This means that while they behave heinously, they remain hard to identify. The underlying cause may be a disconnect between brain regions linking logical decision-making and emotion, leaving them unable to grasp the fallout from their behavior.



MIMICKING EMOTIONS

Posterior cingulate cortex
This region is active when our environment changes and when we are thinking about ourselves. It may help assess the seriousness of offenses and the appropriate response by acting as a hub for integrating intuitions about the mental states of others.



INTERNAL VIEW

Medial frontal gyrus
This region of the brain is important for decision-making and for choosing between alternative potential actions. This is especially the case when there is conflict between multiple options.

SEEING SOMEONE HURT BY ACCIDENT PRODUCES SIMILAR BRAIN ACTIVITY AS IF THE VIEWER WAS HURT THEMSELVES

Orbitofrontal prefrontal cortex
Activated by watching morally charged scenes, this area processes emotional stimuli. It aids in representing just rewards and punishments for observed behavior and in making emotionally driven moral choices.

CAN BRAIN DAMAGE AFFECT MORALITY?

It depends on the area affected. For example, damage to regions that link emotion to moral choice can cause people to make "coldhearted" decisions.